

Section C - Wires and Wire Numbers

[Topics](#) [Send Feedback](#)

Description

Insert wires and wire numbers.

Topics

Name	Description
ace_get_default_wiretype	Get the default wire layer on the active drawing where the wires be inserted when the "Move all lines to wire layers" is checked.
ace_remove_wiretype	Remove the specify layer from wire type on the active drawing. 1 layer is not deleted from the drawing.
ace_set_default_wiretype	Set the default wire type on the active drawing which can be use set the wire layer when inserting a saved circuit.
ace_wnum_find	Test whether the wire number has been used. If the wire numbe use, then return the next unused wire number.
c:ace_del_wnum	Delete wire number and any extra wire number copies found on wire network. This routine also blanks out the connected X?TERM attribute values for component connections tied to the wire netw the wire number is an "in-line" wire number, the wire will "heal" the wire number is removed.
c:ace_fix_wnum	Change existing wire number to either FIXED or NORMAL.
c:ace_get_wiretype_data	Get the full list of data associated with wiretype "layername" on drawing in drawing "fnam" of the active project.
c:ace_get_wiretype_list	Get list of defined wire types (i.e. wire "layers") from target draw file (or active drawing)
c:ace_get_wnum	Retrieve wire number value from selected wire network.
c:ace_insert_wire	Insert wire routing, with auto-routing and collision checking optic wire crosses another wire, AcadE automatically adds a gap or loc crossing point (if defined in dwg's CONFIG dialog box). If it cross one or more devices floating in space, it will break across the components if wire connection points are in the wire's path.
c:ace_mod_wiretype	Modify existing wire type parameter value assignments.
c:ace_new_wiretype	Add new wire layer to active drawing.
c:ace_sig_update	Update cross-referencing on Wire Signals or Stand-Alone referen symbols.
c:ace_toggle_inline	Toggle an in-line wire number to an "above/below" line wire num vice-versa.
c:ace_wnum_fix_wire_cp	Find and fix any gaps of LTYPE wires underneath given wire number b

	point information from component entity "ben".
c:wd_get_wire_netlst	Extract data related to wire netlist. Netlist includes wire segment "wen".
c:wd_inwire_connect_doit	Inserts wiring between components if they have wire connection that are "in line" with each other.
c:wd_move_wn	Move existing wire number entity to a new "XY" coordinate on a entity "line_en" of the same wire network. If the new location do provide enough room, the wire number tries to position itself on leader.
c:wd_putwn	Insert normal (not "fixed") wire number "newval" on wire entity. Inserted above, below, or in-line based upon the drawing's configuration. AcadE automatically positions the wire number on wire segment "en", adding a leader, if necessary, when not in-line and in a congested area. If an existing wire number is found on the segment or anywhere on the network that wire segment "en" is a part of, AcadE updates the existing wire number with value "newval".
c:wd_putwnf	Same as wd_putwn but insert fixed wire number.
c:wd_putwnxy	Same as wd_putwnxyf but inserts a regular wire number.
c:wd_putwnxyf	Put fixed wire number "newval" on LINE that passes through coordinate "xy". AcadE automatically positions the wire number on wire segment/network passing through location "xy". If an existing wire number is found on the segment or anywhere on the network that wire segment "en" is a part of, AcadE updates the existing wire number with value "newval".
c:wd_putwnxyfq	Same as wd_putwnxyf but no error checking and assumes NO existing wire number on network (runs faster in automated drawing generation programs).
c:wd_putwnxyq	Same as wd_putwnxy but no error checking and assumes NO existing wire number on network (runs faster in automated drawing generation programs).
c:wd_strblk_inline_wn	Adjust gap spacing of in-line wire number or in-line wire label.
c:wd_trimwire	Trim wire "ent" at loc "xy". Remove wire connection dot(s) as required.
c:wd_ungap	Attempt to remove any unrequired gap/loops from wire segment
c:wd_wire	Insert single wire segment. If wire crosses another wire, AcadE automatically adds a gap or loop at crossing point (if defined in c:CONFIG dialog box). If it crosses one or more devices floating in space, it will break across the components if wire connection point in the wire's path.
c:wd_wire_num_qm	Change wire number attributes to "?" (this is used prior to a project wide sequential wire numbering operation).
c:wd_wire_quik	Insert wire segment, quick process. Same as wd_wire but with reduced error checking (runs faster in automated drawing generation programs). If wire crosses another wire, AcadE automatically adds gap or loop at crossing point (if defined in dwg's CONFIG dialog box). If it crosses one or more devices floating in space, it will break across the components if wire connection point in the wire's path.

Links

[Topics](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

ace_get_default_wiretype

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Get the default wire layer on the active drawing where the wires will be inserted when the "Move all lines to wire layer" option is checked.

AutoLISP

```
(ace_get_default_wiretype)
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

ace_remove_wiretype

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Remove the specified layer from wire type on the active drawing. The layer is not deleted from the drawing.

AutoLISP

```
(ace_remove_wiretype layertype)
```

Parameters

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

ace_set_default_wiretype

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Set the default wire type on the active drawing which can be used to set the wire layer when inserting a saved circ

AutoLISP

```
(ace_set_default_wiretype layertype)
```

Parameters

Parameters	Description
layertype	The name of the layer to be set

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

ace_wnum_find

[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

```
(ace_wnum_find paramlst)
```

Parameters

Parameters	Description
paramlst	wire number (text string) to test, OR (list option start w_layer dwg_wirenum_lst proj_wirenum_lst) where: "options_lst" = wire number (text string) to test, OR (list option start fmt w_layer dwg_wirenum_lst proj_wirenum_lst) "option" = future "base_start" = start search at this base wire number %N base part of the wire number (nil = WD_M block's WIRE-START attrib value) "fmt" = wire number format (nil = format is just "%N") "w_layer" = optional wire layer name (for wire number on per-layer format) "dwg_wirenum_lst" = optional list of existing wire numbers on active drawing (speed-up)

Example

Check whether wire number "1000" is in use.

```
(setq wnum "1000")  
(setq return (c:ace_wnum_find wnum))  
; If return = "1000" then the wire number is NOT in use  
; If return > "1000" then the return value is the next available  
wire number
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:ace_del_wnum
[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Delete wire number and any extra wire number copies found on target wire network. This routine also blanks out the connected X?TERMxx attribute values for component connections tied to the wire network. If the wire number is an "line" wire number, the wire will "heal" after the wire number is removed.

Parameters

Parameters	Description
en	Entity name of LINE wire of network or of wire num block insert.

Returns

Failure returns nil. Success returns (list wlst) where "wlst" is a list of wire LINE entity segments of the processed wire network.

Example

Remove wire numbers tied to wires marked as 4, 6, or 8 gauge (i.e. layer name assignment is "_4AWG", "_6AWG" or "_8AWG").

```
(defun c:wirenum_remove_large_gauge ( / ss slen ix wnum_txt wire_layername
    ed x wire_ent wnum_ent)
  ; Remove wire numbers tied to wires marked as 4, 6, or 8 gauge (i.e. layer
  ;   name assignment is "_4AWG", "_6AWG", and or "_8AWG").

  ; Quickly find all block inserts that appear to be wire numbers
  (setq ss (ssget "_X" '((-4 . "<AND")(0 . "INSERT")(2 . "WD_WN*")(-4 . "AND>"))))
  (if (/= ss nil)
    (progn ; some found, continue processing
      (setq slen (sslength ss)) ; count of wire number inserts
      (setq ix 0) ; index
      (while (< ix slen)
        (setq wnum_ent (ssname ss ix)) ; get next wire number to process
        ; Get attribute value "WIRENO" or "WIRENOF" off of this insert
        (setq wnum_txt (c:wd_getattrval wnum_ent "WIRENO*"))
        (if wnum_txt ; attribute exists, continue
          (progn
            ; Find LINE wires tied to this wire number
            (setq x (c:ace_wnum_find_wire_en wnum_ent nil))
            (if (AND x (car x))
              (progn ; success, found wire. Check its layer name.
                (setq wire_ent (car x)) ; wire is first element of return
                (setq ed (entget wire_ent))
                ; Layer name of wire LINE entity is subrec 8
                (setq wire_layername (cdr (assoc 8 ed)))
                ; Check for large diameter wiring
                (setq wire_layername (strcase wire_layername)) ; force uppercase
                (if (wcmatch wire_layername "_4AWG,*_6AWG,*_8AWG")
                  (progn ; This wire number is on a larger diameter wire.
                    ; Remove wire number assignment from this wire network.
                    (c:ace_del_wnum wire_ent)
                    (princ "nRemoved wire number: ")
                    (princ wnum_txt)
                    (princ " layer=")
                    (princ wire_layername)
                  )
                )
              )
            )
          )
        (setq ix (1+ ix)) ; increment index to process next wire number
      )
      (setq ss nil) ; release the selection set
    )
    (princ) ; suppress any return
  )
)
```

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:ace_fix_wnum

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Change existing wire number to either FIXED or NORMAL.

AutoLISP

```
(c:ace_fix_wnum en options)
```

Parameters

Parameters	Description
en	entity name of LINE wire or of wire number block IN
options	nil or 1 to change to FIXED, 0 to change to NORMAL

Returns

Failure returns nil. Success returns 1 (even if nothing changed).

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:ace_get_wiretype_data

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

```
(c:ace get wiretype data layername fnam)
```

Parameters	Description
layername	the wire type
fnam	drawing path/file name, drawing name without path, drawing name without ".dwg", or nil=active drawing

nil if not found or wiretype list of data in this format:
(list color size USER1 USER2 ... USER20 wirenumbering)
where "wirenumbering" = "1" if "Yes" and "0" if "No"

Section C - Wires and Wire Numbers

Section C - Wires and Wire Numbers

```
c:ace_get_wiretype_list
```

[Example 1](#) [Example 2](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Get list of defined wire types (i.e. wire "layers") from target drawing file (or active drawing)

```
(c:ace get wiretype list param lst)
```

Parameters

Returns

nil if none found or list of defined wire type names like this: ("BLK_14AWG" "WHT_14AWG" "RED_18AWG")

Example 1

Return list of defined wire types on active drawing.

```
(setq wiretypes (c:ace_get_wiretype_list nil))
```

Example 2

Return list of wire types on specific drawing.

```
(setq wiretypes (c:ace_get_wiretype_list (list "c:/project/schem05.dwg")))
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example 1](#), [Example 2](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:ace_get_wnum
[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Retrieve wire number value from selected wire network.

AutoLISP

```
(c:ace_get_wnum wen)
```

Parameters

Parameters	Description
wen	Entity name of LINE wire segment of network.

Returns

```
(defun c:wn_suffix ( /)
; PURPOSE: insert wire color suffix on to selected wire numbers
(while (setq x (entsel "nSelect LINE object:"))
  (setq wire_ent_name (car x))
  ; Look for AcadE wire number tied to this wire
  (setq return (c:ace_get_wnum wire_ent_name))
  (if (/= return nil)
    (progn
      (setq old_wire_text (car return))
      (setq wire_number_block_ent_name (cadr return))

      ; Get LAYER name of the wire line
      (setq wire_ent_data (entget wire_ent_name))
      (setq wire_layer_name (cdr (assoc 8 wire_ent_data)))

      ; Now look for color substring match in the layer name
      (setq suffix "")
      (cond
        ((wcmatch wire_layer_name "*RED*")
         (setq suffix "RD"))
        )
        ((wcmatch wire_layer_name "*GRN*")
         (setq suffix "GN"))
        )
        ((wcmatch wire_layer_name "*WHT*")
         (setq suffix "WH"))
        )
        ((wcmatch wire_layer_name "*BLK*")
         (setq suffix "BK"))
        )
        ((wcmatch wire_layer_name "*YEL*")
         (setq suffix "YL"))
        )
      )
    (if (/= suffix "")
      (progn ; apply suffix to wire number if it is not already present
        (if (not (wcmatch old_wire_text (strcat "*" suffix)))
          (progn
            (setq new_wire_text (strcat old_wire_text "-" suffix))
            ; Write new "FIXED" wire number out to the wire network
            (c:wd_putwnf wire_ent_name new_wire_text)
          )
          ; ELSE
          (progn
            (princ " color suffix is already present")
          )
        )
      )
    )
  )
)
(princ)
)
```

Section C - Wires and Wire Numbers

Example, Section C - Wires and Wire Numbers

Description

Insert wire routing, with auto-routing and collision checking option. If wire crosses another wire, AcadE automatical a gap or loop at crossing point (if defined in dwg's CONFIG dialog box). If it crosses one or more devices floating in it will break across the components if wire connection points are in the wire's path.

AutoLISP

```
(c:ace_insert_wire param_lst)
```

Parameters

Parameters	Description
param_lst	Flags: nth 0 = starting XY coordinate (can be empty space, from existing LINE wire, or at a component's connection point. nth 1 = ending XY coordinate. nth 2 = "layer" = nil to default to current wire layer current layer not a wire layer then default to first wire layer defined. Otherwise, insert new wire on layer "layer" (automatically created if it doesn't exist). nth 3 = "options" =1's bit set = start and end wires at exact "From" and "To" locs. Don't adjust ends to mate with some wire component connection point. =2's bit set = Do not filter out short line segments. =4's bit set = "auto path calculation" mode is ON. =8's bit set = suppress returning list of inserted wire handles.

Returns

list of inserted wire LINE handles (or nil if options bit 8 is set).

Remarks

Use global **GBL_wd_skip_collision_check** to switch wire collision checking on or off. If **GBL_wd_skip_collision_check** is nil or 0, do wire collision checking. If **GBL_wd_skip_collision_check** is 1 or T wire collision checking.

Example

Route wire from point A to point B.

```
(setq pointA (list 3.5 5.125 0.0))  
(setq pointB (list 10.1 12.25 0.0))  
(c:ace_insert_wire (list pointA pointB "RED_16_AWG" 4))
```

Module

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:ace_mod_wiretype

[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Modify existing wire type parameter value assignments.

AutoLISP

```
(c:ace_mod_wiretype layer_name layer_data options_lst)
```

Parameters

Parameters	Description
layer_name	Layer name to modify (must be existing wire layer)
layer_data	List of parameter and value pairs, ex: (list "COLOR" "SIZE" "14" "USER1" "THHN") where the first entry in list is the layer name itself. Note: the COLOR parameter is a data value for the wire type, it is not used to set the layer's default color value.
options_lst	nil (not used)

Returns

nil if fail or wire layer name if success.

Example

change USER1 and SIZE assignment of wire layer "RED-AWG"

```
(c:ace_mod_wiretype "RED-AWG" (list "SIZE" "14" "USER1" "XHW") nil)
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

c:ace_new_wiretype

[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Add new wire layer to active drawing.

AutoLISP

```
(c:ace_new_wiretype layer_name layer_data options_lst)
```

Parameters

Parameters	Description
layer_name	Layer name to create.
layer_data	List of parameter and value pairs, for example: (list COLOR "RED" SIZE "14" USER1 "THHN"). where the first entry in the list is the layer name itse Note: the COLOR parameter is a data value for the wire type, it is not used to set the layer's default color value. Use "WIRENO" flag with value of "0" to flag the wire as "no wire numbering".
options_lst	nil (not used)

Returns

nil if failure, or new wire layer name if success.

Notes

The WIRE COLOR parameter is a data value for the wire type, it is not used to set the layer's default color value. To the layer defaults, use parameters "_COLOR", "_LTYPE", and "_LWEIGHT".

Example

Create new wire layer "DC-WIRE".

```
(c:ace_new_wiretype "DC-WIRE" (list "WIRE COLOR" "BL" "SIZE" "16AWG" "_COLOR" "BLUE" "_LTYPE" "DASHED" "_LWEIGHT" "2.5") nil)
```

Module

[Section C - Wires and Wire Numbers](#)

What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:ace_sig_update
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Update cross-referencing on Wire Signals or Stand-Alone reference symbols.

AutoLISP

```
(c:ace_sig_update param_lst)
```

Parameters

Parameters	Description
param_lst	(list dowhat dowir doref doshow entlst options) wher dowhat = 1 to process wire signals, = 0 to process s alone cross-ref symbols dowir = (if dowhat=1) 1 to process wire nums tied t signals; 0=skip doref = 1 to update cross-refs on signals; 0= skip doshow = 1 to show src/dst links with temp graphics 0=skip entlst = list of entities to process; = nil to process ei dwg options = future

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:ace_toggle_inline
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Toaqle an in-line wire number to an "above/below" line wire number or vice-versa.

Parameters

Parameters	Description
en	Entity name of LINE wire or the wire number block INSERT.
options	For future use.

Returns

Nil if no action, otherwise returns handle of the toggled wire number block insert.

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:ace_wnum_find_wire_en

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Find entity name of LINE wire underneath given wire number block insert.

AutoLISP

```
(c:ace_wnum_find_wire_en wnum_en trap)
```

Parameters

Parameters	Description
wnum_en	Wire number block insert.
trap	trap distance to search outward from wire number block insert's XY insertion point. If nil then default to draw default trap distance given in global GBL_wd_trp.

"line_lst" = list of all LINE entities found under the wire number insert (success if only one and it is the same as "wire_en").

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference

c:wd_do_str_wire
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Stretch wire to try to connect to a wire connection point.

AutoLISP

```
(c:wd_do_str_wire wire_en loc)
```

Parameters

Parameters	Description
wire_en	Entity name of WIRE line segment.
loc	Identify which end of LINE to stretch.

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference

c:wd_get_sym_pntlst

Extract list of lists of wire connection point information. Get connection point information from component entity "b

AutoLISP

```
(c:wd_get_sym_pntlst ben nozoom usecache)
```

Parameters

Parameters	Description
ben	Entity name of component (block insert with attribut
nozoom	1 to suppress zoom ext, otherwise allow zoom ext if symbol not entirely on screen.
usecache	1 to look for target "ben" in existing cache.

Returns

List of lists, one list for each wire connection point carried on the target entity. For example: (list (list index connection_xyz nil conn_wire_en_lst suffix xterm_ent_name multi_wire_lst maxcon jumpers_attr_en) (list index... Where :

index = index number (integer).

connection_xyz = XYZ coordinate of the X?TERMxx wire connection attribute.

nil = for future use.

conn_wire_en_lst = list of entity names of LINE segments that have end points that touch or "connect to" the wire connection point.

xdir = the "?" character of the X?TERMxx wire connection attribute (valid values = 0, 1,2,4, or 8).

suffix = the "xx" suffix characters of the X?TERMxx wire connection attribute (ex: "01").

xterm_en_name = entity name of the X?TERMxx wire connection attribute.

multi_wire_lst = list of entity names of polylines on the "_MULTI_WIRE" layer that tie into the wire connection point

maxcon = (future) value of "PREFER_MAXCON" xdata carried on wire connection attribute.

jumpers_attr_en = (may not be present in the list) - entity name of WD_JUMPERS attribute (if exists, otherwise thi is omitted from the list).

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Description

Extract data related to wire netlist. Netlist includes wire segment entity "wen".

AutoLISP

```
(c:wd_get_wire_netlst wen options)
```

Parameters

Parameters	Description
wen	AutoCAD entity name of line wire segment.
options	nil = no options. Bit 1 = 0 to ignore all terminal symbols that don't ch wire num and read through symbol screw terms whe mult wire connection points on screw term. Bit 1 = 1 (for various wire connection reports). All terminals treated same as components. Break wire a don't read through a symbol's screw term with mult. Bit 2 = 1 suppress extract of component and X?TERI attributes. Bit 4 = 1 to use GBL_wd_ecache and GBL_wd_dcach "pntlst" cache. Bit 8 = 1 to not read through CABLE (?W0*) marke Bit 16 = single-step debug trace mode. Bit 32 = 1 for quick CABLE marker search on networ Bit 64 = 1 to also process LINE entities on "multi-wir line layer. Bit 128 = 1 to force PNLWDLEV query or search thru on current dwg for WDLEV values.

Returns

List of lists: (list wlst en1st xmin ymax s_sig d_sig trmlst xtermlst cmplst cmpconlst fan_inout_lst nil) where:

wlst = list of wire LINE segment entity names that make up the network.

en1st = entity name of upper left-most "beginning" wire segment of the network.

xmin =

ymax =

s_sig =

d_sig =

trmlst =

xtermlst =

cmplst =

cmpconlst = list of connected component connection data lists (list (list comp_en xterm_en wire_en cbl_en wireseq
future prefer_maxcon) ...) where:

wireseqval = optional wire sequence code assigned to "xterm_en", default="900".

prefer_maxcon = optional max wire connection value tied to "xterm_en".

fan_inout_lst = list of any fan in/out symbols.

Example

This snippet of code uses the wire "netlist" to find the component tags at each end of a wire and then calculate a sp wire number based upon this information.

```
(defun c:milnum ( / en ed x compconlst comp1_en comp2_en pin1 pin2 new_wireno netlst str comp1_
comp2_tag
xterm1_en xterm2_en)
  (while (setq x (entsel "nSelect wire for special MIL wire number format:"))
    (setq en (car x))
    (setq ed (entget en))
    (if (= (cdr (assoc 0 ed)) "LINE")
      (progn ; okay to continue
        (if (setq netlst (c:wd_get_wire_netlst en 1))
          (progn
            (setq compconlst (nth 9 netlst))
            (cond
              ((= (length compconlst) 2)
               ; two connected components, okay to continue
               (setq comp1_en (car (car compconlst)))
               (setq comp2_en (car (cadr compconlst)))
               (setq xterm1_en (cadr (car compconlst)))
               (setq xterm2_en (cadr (cadr compconlst)))
               ; Get the suffix of the first component's X?TERMxx attrib
               (setq str (cdr (assoc 2 (entget xterm1_en))))
               ; Get PIN number of first component
               (setq pin1 (c:wd_getattrval comp1_en (strcat "TERM" (substr str 7))))
               (if (AND (not pin1) (c:wd_getattrval comp1_en "TAGSTRIP"))
                 ; Terminal symbol, just grab the TERM01 terminal number
                 (setq pin1 (c:wd_getattrval comp1_en "TERM01")))
               ; Get the suffix of the 2nd component's X?TERMxx attrib
               (setq str (cdr (assoc 2 (entget xterm2_en))))
               ; Get PIN number of first component
               (setq pin2 (c:wd_getattrval comp2_en (strcat "TERM" (substr str 7))))
               (if (AND (not pin2) (c:wd_getattrval comp2_en "TAGSTRIP"))
                 ; Terminal symbol, just grab the TERM01 terminal number
                 (setq pin2 (c:wd_getattrval comp2_en "TERM01")))
               ; Get tag of the two components
               (setq comp1_tag (c:wd_getattrval comp1_en "TAG1*,TAG2*,TAGSTRIP,TAG,TAG_*"))
               (setq comp2_tag (c:wd_getattrval comp2_en "TAG1*,TAG2*,TAGSTRIP,TAG,TAG_*"))
               (if (not comp1_tag) (setq comp1_tag ""))
               (if (not comp2_tag) (setq comp2_tag ""))
               (setq new_wireno comp1_tag)
               (if (AND pin1 (/= pin1 "")) (setq new_wireno (strcat new_wireno "-" pin1)))
               (setq new_wireno (strcat new_wireno "/" comp2_tag))
               (if (AND pin2 (/= pin2 "")) (setq new_wireno (strcat new_wireno "-" pin2)))
               (c:wd_putwnf en new_wireno) ; put the wire number on wire
              )
            )
          (T (princ "n other than two connected componentsn")))
        )
      )
    )
  )
  (princ)
)
```

Module

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:wd_inwire_connect_doit
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Inserts wiring between components if they have wire connection points that are "in line" with each other.

AutoLISP

```
(c:wd_inwire_connect_doit ben1 ben2 layer)
```

Parameters

Parameters	Description
ben1	AutoCAD entity name (or handle) of first component
ben2	AutoCAD entity name (or handle) of second component
layer	nil to default to current wire layer or, if current layer wire layer then default to first wire layer defined. Otherwise, insert new wire on layer "layer".

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:wd_move_wn
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Move existing wire number entity to a new "XY" coordinate on a line entity "line_en" of the same wire network. If the

```
(c:wd_move_wn old_ben line_en xy)
```

Parameters

Parameters	Description
old_ben	Wire number block name.
line_en	Ent name of wire segment to move wire number to.
xy	xy coordinate on the line where wire number to mov

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference

c:wd_putwn
[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Insert normal (not "fixed") wire number "newval" on wire entity "en". Inserted above, below, or in-line based upon drawing's configuration. AcadE automatically positions the wire number on or in wire segment "en", adding a lead necessary, when not in-line and in a congested area. If an existing wire number is found on the segment or anywhere the network that wire segment "en" is a part of, AcadE updates the existing wire number with value "newval".

AutoLISP

```
(c:wd_putwn en newval)
```

Parameters

Parameters	Description
en	AutoCAD entity name of LINE wire.
newval	new wire number value.

Returns

```
(while (setq pick (entsel "npick wire for SPARE:")) (c:wd_putwn (car pick) "SPARE"))
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference

c:wd_putwnf
[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Same as wd_putwn but insert fixed wire number.

AutoLISP

```
(c:wd_putwnf en newval)
```

Parameters

Parameters	Description
en	AutoCAD entity name of LINE wire.
newval	new wire number value.

Example

This snippet of code uses the (c:wd_putwnf) function to manually insert incrementing PLC I/O address-based wire numbers.

```
(defun c:plcwnum ( / en ed x new_wnum base_wnum ix_cnt index pt1 pt2 wnum_blk_hdl pick_xy xitno
suffixlst
new_wireno ins_xy)
; Semi-automatically insert incrementing I/O address wire numbers
(setq suffixlst (list "A" "B" "C" "D" "E" "F" "G" "H" "J" "K" "L" "M" "N" "P" "R"))
(while (setq x (entsel "nSelect MAIN I/O wire connection (LINE wire, WNUM, or I/O address
text):"))
  (setq ix_cnt "")
  (setq base_wnum nil)
  (setq en (car x))
  (setq pick_xy (cadr x)) ; point point
  (setq ed (entget en))
  (cond
    ((= (cdr (assoc 0 ed)) "INCPRT") ; user may have picked on wire number block
```

```

(setq en (car x)) ; picked entity
(setq ed (entget en))
(if (AND (= (cdr (assoc 0 ed)) "ATTRIB")
        (wcmatch (cdr (assoc 2 ed)) "TAGA*")))
  (progn ; Yes, got it. Get I/O address value TAGA* attribute value
    (setq base_wnum (cdr (assoc 1 ed)))
  ))))
)
(= (cdr (assoc 0 ed)) "LINE") ; user probably picked on line WIRE
; Look for wire number pointer tied to this LINE entity. Points at the
; wire number block insert.
(setq wnum_blk_hdl (car (wd_wnet_find_ptr_for_whole_net en)))
(if (AND wnum_blk_hdl (handent wnum_blk_hdl))
    (progn
      (setq base_wnum (c:wd_getattrval (handent wnum_blk_hdl) "WIRENO*"))
    )
  )
)
)
)
)
(if base_wnum
  (progn ; Okay, now prompt user to pick on wires for the incrementing
    ; I/O address-based wire number assignments.
    (setq index 0) ; start at beginning of suffix list
    (setq new_wireno (strcat base_wnum (nth index suffixlst)))
    (setq xitnow nil)
    (while (AND (not xitnow)
      (setq x (entsel (strcat "n Pick wire for wnum assignment: " new_wireno))))
      (setq en (car x))
      (setq ed (entget en))
      (cond
        ((= (cdr (assoc 0 ed)) "LINE") ; user picked on a "WIRE" (hopefully!)
          (c:wd_putwnf en new_wireno) ; insert wire number on selected network
        )
        ((= (cdr (assoc 0 ed)) "INSERT") ; user picked on a block insert (!)
          ; See if lucky and user picked on a wire number block insert. Block
          ; name will have a "WD_WN" prefix if it's a wire number block insert.
          (if (wcmatch (cdr (assoc 2 ed)) "WD_WN*")
            (progn ; yes, user picked the wire number insert. Need to find the LINE
              ; wire that it's sitting on.
              (setq ins_xy (cdr (assoc 10 ed))) ; XY coord of wire number block insert
              ; Set up a diagonal fence to cut across and capture any underlying
              ; line wire.
              (setq pt1 (list (- (car ins_xy) 0.05) (- (cadr ins_xy) 0.05) 0.0))
              (setq pt2 (list (+ (car ins_xy) 0.05) (+ (cadr ins_xy) 0.05) 0.0))
              (setq ss (ssget "_F" (list pt1 pt2) '((0 . "LINE"))))
              (if (/= ss nil)
                (progn ; a LINE entity captured.
                  (setq en (ssname ss 0)) ; first or only LINE captured
                  (setq ss nil) ; release the selection set
                  (c:wd_putwnf en new_wireno) ; pop in the wire number (it should
                    ; update the existing one and reset the component wire
                    ; connection X?TERMxx attributes that "touch" this wire.
                  ))
              ))
          )
        )
      )
    )
  )
  (setq index (1+ index)) ; increment pointer
  (if (= index (length suffixlst))
    (progn ; no more suffixes remain, exit now
      (setq xitnow 1)
    )
    ; ELSE
    (progn ; calculate next wire number
      (setq new_wireno (strcat base_wnum (nth index suffixlst)))
    )
  )
  ; If AcadE had to do a ZOOM extents to follow the wire number off-screen
  ; then variable "Z_prev" will be set to 1. If so, do a ZOOM PREV to return
  ; the display to the original position.

```

)

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference

c:wd_putwnxy

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Same as wd_putwnxyf but inserts a regular wire number.

AutoLISP

```
(c:wd_putwnxy xy newval)
```

Parameters

Parameters	Description
xy	Point that the line passes through.
newval	New wire number value.

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:wd_putwnxyf
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

number on the wire segment/network passing through location "xy". If an existing wire number is found on the segment or anywhere on the network that wire segment "en" is a part of, AcadE updates the existing wire number with value "newval".

AutoLISP

```
(c:wd_putwnxyf xy newval)
```

Parameters

Parameters	Description
xy	Point that the line passes through.
newval	New wire number value.

Returns

Ent name of the wire number block insert.

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:wd_putwnxyfq
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Same as wd_putwnxyf but no error checking and assumes NO existing wire number on network (runs faster in automatic drawing generation programs).

AutoLISP

```
(c:wd_putwnxyfq xy newval)
```

Parameters

Parameters	Description
------------	-------------

Remarks

DO NOT use this routine if there could be an existing wire number present on the wire network; use wd_putwnxy in

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:wd_putwnxyq
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Same as wd_putwnxy but no error checking and assumes NO existing wire number on network (runs faster in autorouting drawing generation programs).

AutoLISP

```
(c:wd_putwnxyq xy newval)
```

Parameters

Parameters	Description
xy	Point that the line passes through.
newval	New wire number value.

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Description

Adjust gap spacing of in-line wire number or in-line wire label.

AutoLISP

```
(c:wd_strblk_inline_wn en param_lst)
```

Parameters

Parameters	Description
en	Entity name of the in-line wire number or in-line wire label.
param_lst	(list min_width text_to_line_gap snapx) where: "min_width" = 0.0 to minimize, <0.0 minimum total width "text_to_line_gap" = minimum gap between edge of and start of wire line "snapx" = snap increments for gap width change, 0. nil = continuous adjustment.

Example

```
(c:wd_strblk_inline_wn (car (entsel)) (list 0.0 0.125 0.0))
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

Description

Trim wire "ent" at loc "xy". Remove wire connection dot(s) as required.

Parameters

Parameters	Description
ent	Entity name of wire to trim.
xy	Location for trim point.

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.
What do you think about this topic? [Send feedback!](#)
AutoCAD Electrical 2025: AutoLISP Reference
c:wd_ungap
[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Attempt to remove any unrequired gap/loops from wire segment.

AutoLISP

(c:wd_ungap wen)

Parameters

Parameters	Description
wen	AutoCAD entity name of wire segment to process.

Returns

nil if no gaps removed, otherwise returns handle of the modified wire segment.

Module

[Section C - Wires and Wire Numbers](#)

■ ■ ■ ■ ■

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:wd_wire

[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Insert single wire segment. If wire crosses another wire, AcadE automatically adds a gap or loop at crossing point (defined in dwg's CONFIG dialog box). If it crosses one or more devices floating in space, it will break across the components if wire connection points are in the wire's path.

AutoLISP

```
(c:wd_wire end1 end2 layer)
```

Parameters

Parameters	Description
end1	Start coordinates.
end2	End coordinates.
layer	nil to default to current wire layer or, if current layer wire layer then default to first wire layer defined. Otherwise, insert new wire on layer "layer" (layer automatically created if it doesn't exist).

Example

```
(c:wd_wire (list 9.5 8.125) (list 12.5 8.125) "RED_14_THHN")
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:wd_wire_num_qm

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

AutoLISP

```
(c:wd_wire_num_qm)
```

Module

[Section C - Wires and Wire Numbers](#)

Links

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:wd_wire_quik

[Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Insert wire segment, quick process. Same as [wd_wire](#) but with reduced error checking (runs faster in automated dr generation programs). If wire crosses another wire, AcadE automatically adds a gap or loop at crossing point (if def dwg's CONFIG dialog box). If it crosses one or more devices floating in space, it will break across the components if connection points are in the wire's path.

AutoLISP

```
(c:wd_wire_quik end1 end2 layer)
```

Parameters

Parameters	Description
end1	Start coordinates.
end2	End coordinates.
layer	nil to default to current wire layer or, if current layer wire layer then default to first wire layer defined. Otherwise, insert new wire on layer "layer" (layer automatically created if it doesn't exist).

Module

[Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)

AutoCAD Electrical 2025: AutoLISP Reference

c:wd_wnums_doit

[Example](#) [Section C - Wires and Wire Numbers](#) [Send Feedback](#)

Description

Project-wide wire number tagging batch process.

AutoLISP

```
(c:wd_wnums_doit dlist freshen param_lst)
```

Parameters

Parameters	Description
dlist	List of dwgs to process (full file names, double back slashes) (nil = process all dwgs in active project set)
freshen	"1" to force database freshen, "0" = no freshen.
param_lst	(nth 0) = "0" (nth 1) = "1" to recalc/overwrite old wire nums, "0" process, untagged wires only (skip existing wire numbers). (nth 2) = starting wire num text string (sequential n only). (nth 3) = wire number format (override value). (nth 4) = "1" to trigger SIGNAL arrow cross-ref processing, "0" to not do SIGNAL cross-referencing update. (nth 5) = "C" = consecutive sequential from one dra to the next (sequential mode only), "S" = use each c CONFIG "starting wire num" (for sequential mode on "R" = reference based tagging mode. (nth 6) = "1" to run prelim pass to erase all existing nums, "0" to not make prelim pass to erase existing nums. (nth 7) = "" (nth 8) = nil or "0" for Normal wire numbers, "1" for wire numbers.

Example

```
(c:wd_wnums_doit nil "1" (list "0" "1" "1" "%N" "1" "R" "0" "" "0"))
```

Module

[Example](#), [Section C - Wires and Wire Numbers](#)

Copyright (c) Autodesk 2024. All rights reserved.

What do you think about this topic? [Send feedback!](#)